Autumn 2025
IIT Bombay

MA 403 Real Analysis
Homework 8

1. In this exercise we show that $f'(c) > 0$ at some point c is not sufficient to guarantee the existence of an open interval containing c on which f is increasing. Consider the function $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = \begin{cases} x + 2x^2 \sin\left(\frac{1}{x}\right) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0. \end{cases}$

(a) Compute $f'(0)$ and $f'(x)$ for $x \neq 0$.

(b) Prove that there exists a sequence of points $\{x_n\}$ with $x_n \neq 0$, $x_n \to 0$ and $f'(x_n) < 0$.

2. In class, we proved that if $f : (a, b) \to \mathbb{R}$ is r^{th} order differentiable at x, and $P(h) := \sum_{k=0}^{r} \frac{f^{(k)}(x)}{k!} h^k$, then $R(h) := f(x + h) - P(h)$ satisfies the property: $\lim_{h\to 0} \frac{R(h)}{h^r} = 0$.

Prove that P is the unique polynomial of degree $\leqslant r$ with this property.

3. (a) Suppose f is defined in an open interval containing a, and suppose $f''(a)$ exists. Show that

$$f''(a) = \lim_{h\to 0} \frac{f(a + h) - 2f(a) + f(a - h)}{h^2}$$

Hint: Taylor's theorem.

(b) Give an example where the limit of the quotient in part (a) exists but where $f''(a)$ does not exist.

4. Locate and classify the points of discontinuity of the following functions f defined on $\mathbb{R}$:

(a) $f(x) = \begin{cases} \frac{\sin x}{x} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0. \end{cases}$

(b) $f(x) = \begin{cases} e^{\frac{1}{x}} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0. \end{cases}$

5. Let f be an increasing function defined on $[a, b]$ and let $x_1, \cdots, x_n$ be n points in (a, b) such that $a < x_1 < x_2 < \cdots < x_n < b$.

(a) Show that $\sum_{k=1}^{n} [f(x_k+) - f(x_k-)] \leqslant f(b) - f(a)$.

(b) For each $m \in \mathbb{Z}_+$, let S_m be the set of points in $[a, b]$ where the jump of f is greater than $\frac{1}{m}$. Use part (a) to show that S_m is a finite set.

(c) Show that the set of discontinuities of f is countable.